Polynomiality of the Lecture Hall sequence via signed Dyck paths

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Abstract

Katthän defined a sequence of rational functions by prescribing certain Toeplitz minors and conjectured that every term is a polynomial. We identify this sequence with a specialization of the Stieltjes–Rogers moment polynomials: the down-step weight at height h is $y_h y_{h+1}$, with a negative sign precisely when $h \geq 4$ is even. Classical Hankel determinant identities, together with the signs arising from row reversal and the uniqueness of the defining recurrence, establish this identification for every index. Thus the polynomiality conjecture follows. We give a self-contained derivation of the required classical identities and an explicit coefficient formula indexed by compositions. The latter implies that the ith term has exactly 2^{i-2} nonzero monomials for $i \geq 2$. We state the interpretation of two typographical inconsistencies in the source and explain its support within that paper.

1 The sequence and its moment representation

Let $R = \mathbb{Z}[y_1, y_2, \ldots]$, where the variables are algebraically independent, and let K be its fraction field. For a sequence $P = (P_1, P_2, \ldots)$ in K, define

$$M(P)_{r,c} = \begin{cases} -P_{c-r+1}, & c \geq r, \\ 0, & c < r, \end{cases} \qquad r, c \geq 1.$$

We use the following square minors:

$$E_i(P) = -\det M(P)_{\{1,\ldots,\lceil i/2 \rceil\},\{\lfloor i/2 \rfloor + 1,\ldots,i\}}, \qquad i \geq 1. \tag{1}$$

Set

$$T_i = \prod_{j=1}^{i} y_j^{i+1-j}. \tag{2}$$

The Lecture Hall sequence is the unique sequence $\ell = (\ell_1, \ell_2, \ldots)$ in K satisfying

$$E_i(\ell) = T_i \qquad (i \geq 1). \tag{3}$$

Existence and uniqueness are recalled in Lemma 3.1 below. Katthän [1, Definition 3.1 and Proposition 3.3] established that these terms are integer Laurent polynomials and conjectured that they are polynomials [1, Conjecture 3.4].

1.1 Interpretation of the source definition

Equation (1) uses the definition determined by the square-matrix description and examples in [1]. On page 3 of the specified arXiv version, both the displayed column set and the following sentence start at $\lfloor i/2 \rfloor$, without the $+1$. That set has one more column than the prescribed

number of rows; for $i = 1$ it also includes column 0. It therefore does not specify an ordinary determinant.

The intended square minor is fixed by three pieces of internal evidence. First, the source describes the largest top-aligned square block, with top-right entry $-P_i$, that contains none of the structural zeros below the diagonal. If its size is k, its columns are $i-k+1, \dots, i$, and the condition is $i-k+1 \geq k$. Its largest possible size is $k = \lceil i/2 \rceil$, giving exactly (1). Second, all four determinants in Example 3.2 agree with (1). Third, Definition 4.1 defines ℓ_S using columns $S+1$ and identifies $\ell_{\{\lfloor i/2 \rfloor, \dots, i-1\}}$ with $E_i(\ell)$. This again gives the columns in (1).

There is a second inconsistency in Example 3.2 on page 4: the printed last term of ℓ_4 is $y_1 y_2^2 y_3^2 y_4^2$. The fourth defining equation, $\ell_2 \ell_4 - \ell_3^2 = T_4$, instead gives

$$\ell_4 = y_1^2 y_2^3 (y_1 + y_3)^2 + y_1^2 y_2^2 y_3^2 y_4. \tag{4}$$

This is also the expression listed in Section 6, page 8, of the source. We use the defining equations and this consistent expression. These are explicit interpretations supported by the source itself; they are not presented as a formally issued erratum.

1.2 Statement and relation to existing theory

A Dyck path of semilength n starts at $(0,0)$, ends at $(2n,0)$, uses steps $(1,1)$ and $(1,-1)$, and never goes below height 0. Give each up step weight 1 and each down step from height h weight α_h. Write $\mu_n(\alpha)$ for the sum of the products of step weights over all such paths, with $\mu_0 = 1$. These are the Stieltjes–Rogers polynomials; see [2, Theorem 9.1(b)] and the path interpretation of continued fractions in [3].

Define the signs and specialized weights by

$$\varepsilon_h = \begin{cases} -1, & h \geq 4 \text{ and } h \text{ is even}, \\ 1, & \text{otherwise}, \end{cases} \qquad \alpha_h = \varepsilon_h y_h y_{h+1}. \tag{5}$$

Theorem 1.1. *For the Lecture Hall sequence defined by* (1)–(3)*,*

$$\ell_{n+1} = y_1 \mu_n(\alpha) \qquad (n \geq 0), \tag{6}$$

with the specialization (5)*. In particular,* $\ell_i \in \mathbb{Z}[y_1, \dots, y_i]$ *for every* $i \geq 1$*.*

This proves Conjecture 3.4 of [1] under the source interpretation in Section 1.1. The contribution presented here is the explicit identification (6) and its connection to the prescribed Toeplitz minors. The underlying Hankel identities are classical: they follow directly from [2, Theorem 9.5(b,c)]. Section 2 includes a derivation to keep the argument self-contained. We do not claim these general identities as new, nor assert priority for the specific identification. The initial-algebra and SAGBI statements in Conjecture 4.2 of [1] are outside the scope of this note.

2 Classical Hankel identities for Dyck moments

In this section the α_h are independent variables. All identities are polynomial identities over $\mathbb{Z}$, so they remain valid under signed specializations, including those with zero weights.

Lemma 2.1 (Hankel products)**.** *For every* $m \geq 1$*, the Dyck moments satisfy*

$$H_m^{(0)} := \det[\mu_{i+j}]_{0 \leq i,j < m} = \prod_{h=1}^{m-1} (\alpha_{2h-1}\alpha_{2h})^{m-h}, \tag{7}$$

$$H_m^{(1)} := \det[\mu_{i+j+1}]_{0 \leq i,j < m} = \alpha_1^m \prod_{h=1}^{m-1} (\alpha_{2h}\alpha_{2h+1})^{m-h}. \tag{8}$$

Empty products are 1.

Proof. Group the steps of a Dyck path into consecutive pairs and divide its heights at even times by 2. The resulting Motzkin path has up-step weight 1 and down-step weight

$$\lambda_h = \alpha_{2h-1}\alpha_{2h} \qquad (h \geq 1).$$

The two possible horizontal steps at height $h \geq 1$ have weights α_{2h} and α_{2h+1}; at height 0 only the latter is allowed. Their total weights are therefore

$$b_0 = \alpha_1, \qquad b_h = \alpha_{2h} + \alpha_{2h+1} \quad (h \geq 1).$$

Let J be the infinite matrix indexed by the nonnegative integers with $J_{h,h} = b_h$ and $J_{h,h+1} = 1$ for $h \geq 0$, and $J_{h,h-1} = \lambda_h$ for $h \geq 1$; all other entries are zero. Matrix multiplication enumerates these paths, so $\mu_n = (J^n)_{0,0}$. Each relevant entry of a fixed power involves only finitely many indices.

Set $d_0 = 1$, $d_h = \lambda_1 \cdots \lambda_h$, and $D = \operatorname{diag}(d_0, d_1, \ldots)$. Directly, $JD = DJ^{\mathsf{T}}$, and hence

$$(J^j)_{h,0} = d_h (J^j)_{0,h} \qquad (h, j \geq 0). \tag{9}$$

No division by d_h is used. Fix $m \geq 1$ and put $L_{i,h} = (J^i)_{0,h}$ for $0 \leq i, h < m$. Since a path of length i cannot reach a height greater than i, and the unique path reaching height i has weight 1, L is unit lower triangular.

Let D_m and J_m denote the leading $m \times m$ blocks of D and J. Using (9) gives

$$[\mu_{i+j}]_{0 \leq i,j < m} = L D_m L^{\mathsf{T}}. \tag{10}$$

Indeed, the intermediate height in $J^i J^j$ is at most $\min(i, j)$. For the shifted matrix, write

$$\begin{aligned} \mu_{i+j+1} &= \sum_{a,b \geq 0} (J^i)_{0,a} J_{a,b} (J^j)_{b,0} \\ &= \sum_{a,b=0}^{m-1} L_{i,a} (J_m)_{a,b} d_b L_{j,b}. \end{aligned}$$

The truncation is exact: a nonzero summand requires $a \leq i < m$ and $b \leq j < m$. Consequently,

$$[\mu_{i+j+1}]_{0 \leq i,j < m} = L J_m D_m L^{\mathsf{T}}. \tag{11}$$

Taking determinants in (10) proves (7).

For (8), let $\delta_m = \det J_m$ and $\delta_0 = 1$. Expansion of a tridiagonal determinant yields

$$\delta_1 = \alpha_1, \qquad \delta_m = b_{m-1}\delta_{m-1} - \lambda_{m-1}\delta_{m-2} \quad (m \geq 2).$$

Induction gives $\delta_m = \prod_{h=1}^{m} \alpha_{2h-1}$: after using the induction hypothesis, the summand containing α_{2m-2} from b_{m-1} cancels the second term of the recurrence. This is a polynomial cancellation and does not require any factor to be invertible. Taking determinants in (11) now gives

$$H_m^{(1)} = \left(\prod_{h=1}^{m} \alpha_{2h-1} \right) \prod_{h=1}^{m-1} (\alpha_{2h-1}\alpha_{2h})^{m-h},$$

which rearranges to (8). For $m = 1$, the two formulas read $H_1^{(0)} = 1$ and $H_1^{(1)} = \alpha_1$, as required. □

For comparison, the two Stieltjes–Rogers factorizations in [2, Theorem 9.5(b,c)] have diagonal entries $\prod_{j=1}^{2h} \alpha_j$ and $\prod_{j=1}^{2h+1} \alpha_j$, respectively, for $h \geq 0$. Their leading principal determinants give exactly (7) and (8). Positivity assumptions from analytic moment problems are not needed here.

3 Identification with the Lecture Hall sequence

Lemma 3.1 (Triangular recurrence)**.** *There is exactly one sequence in K satisfying* (3)*.*

Proof. The first two equations give $\ell_1 = y_1$ and $\ell_2 = y_1^2 y_2$. For $i \geq 3$, put $k = \lceil i/2 \rceil$. The entry $-P_i$ occurs just once in the minor in (1), in its first row and last column. Its cofactor, after shifting both remaining index sets down by 1, is the minor defining E_{i-2}. Keeping track of the determinant signs gives

$$E_i(P) = (-1)^k E_{i-2}(P) P_i + Q_i(P_1, \ldots, P_{i-1})$$

for a polynomial Q_i. After the previous equations have been solved, the coefficient of P_i is the nonzero monomial $(-1)^k T_{i-2}$. It is invertible in K, so each equation has exactly one solution for its new term. Induction proves both existence and uniqueness. □

Proof of Theorem 1.1. Let $P_{n+1} = y_1 \mu_n(\alpha)$ with (5). Define

$$s_m = (-1)^{1+m(m+1)/2}.$$

Reverse the row order in the square minor (1) and factor a minus sign from each row. The row reversal has sign $(-1)^{m(m-1)/2}$, and the negative sign outside the determinant must also be included. Thus for every $m \geq 1$,

$$E_{2m-1}(P) = s_m \det[P_{r+s+1}]_{0 \leq r,s < m} = s_m y_1^m H_m^{(0)}, \tag{12}$$

$$E_{2m}(P) = s_m \det[P_{r+s+2}]_{0 \leq r,s < m} = s_m y_1^m H_m^{(1)}. \tag{13}$$

Apply Lemma 2.1. In (7), both α_{2h-1} and α_{2h} have exponent $m-h$. After the substitution (5) and multiplication by y_1^m, the exponent of y_j is $2m-j$ for $1 \leq j \leq 2m-1$. In (8), α_1 has exponent m, while α_{2h} and α_{2h+1} have exponent $m-h$. The resulting exponent of y_j is $2m+1-j$ for $1 \leq j \leq 2m$. These are precisely the exponents in T_{2m-1} and T_{2m}.

For $m \geq 2$, the sign in each product is

$$(-1)^{\sum_{h=2}^{m-1}(m-h)} = (-1)^{(m-2)(m-1)/2} = s_m,$$

since the last exponent and $1 + m(m+1)/2$ differ by $2m$. For $m = 1$ the two products give y_1 and $y_1^2 y_2$ directly, and $s_1 = 1$. Hence

$$y_1^m H_m^{(0)} = s_m T_{2m-1}, \qquad y_1^m H_m^{(1)} = s_m T_{2m}.$$

Equations (12)–(13) therefore give $E_i(P) = T_i$ for every $i \geq 1$. Lemma 3.1 implies $P_i = \ell_i$. Finally, a Dyck path of semilength n has height at most n, so $y_1 \mu_n(\alpha)$ is a finite sum in $\mathbb{Z}[y_1, \ldots, y_{n+1}]$. □

Remark 3.2. The uniqueness argument takes place in the fraction field of independent variables. This stepwise division argument cannot be applied directly to specializations at which a divisor vanishes. This does not restrict the polynomial identity in Theorem 1.1, which can itself be evaluated at arbitrary values.

4 Coefficients and generating function

Proposition 4.1. *For every $n \geq 1$,*

$$\ell_{n+1} = y_1 \sum_{\substack{r \geq 1,\ m_1 + \cdots + m_r = n \\ m_1, \ldots, m_r \geq 1}} \left(\prod_{j=1}^{r-1} \binom{m_j + m_{j+1} - 1}{m_{j+1}} \right) (-1)^{m_4 + m_6 + \cdots} \prod_{j=1}^{r} (y_j y_{j+1})^{m_j}. \tag{14}$$

An absent m_j is read as zero. Consequently ℓ_i has exactly 2^{i-2} nonzero monomials for every $i \geq 2$.

Proof. For a nonempty Dyck path, let m_j be the number of down steps from height j, and let r be its maximum height. Every m_j with $1 \le j \le r$ is positive and their sum is n.

Use the usual contour correspondence between Dyck paths and plane rooted trees. Then m_j is the number of vertices at depth j. Given the vertices at depth j in their left-to-right order, specifying the number of children of each is a weak composition of m_{j+1} into m_j parts. There are $\binom{m_j+m_{j+1}-1}{m_{j+1}}$ choices. These choices can be made independently at each depth and determine the ordered tree uniquely. The root has m_1 children, contributing a factor of 1. Thus the product of binomial coefficients in (14) counts paths with the prescribed profile. Their common weight is $\prod_j \alpha_j^{m_j}$, which proves the formula by Theorem 1.1.

Different compositions produce different monomials. If e_j is the exponent of y_j, then $m_1 = e_1 - 1$ and $m_j = e_j - m_{j-1}$ for $j \ge 2$, with missing parts set to zero. Thus the profile can be recovered from the monomial. Each coefficient has strictly positive absolute value, so no profile disappears by cancellation. There are 2^{n-1} compositions of n, giving the claim with $i = n + 1$. □

For example, the first terms are $\ell_1 = y_1$, $\ell_2 = y_1^2 y_2$, $\ell_3 = y_1^2 y_2^2 (y_1 + y_3)$, the expression (4), and

$$\begin{aligned}\ell_5 = {} & y_1^2 y_2^4 (y_1 + y_3)^3 + 2 y_1^2 y_2^3 y_3^2 y_4 (y_1 + y_3) \\ & + y_1^2 y_2^2 y_3^2 y_4^2 (y_3 - y_5).\end{aligned}$$

In particular, polynomiality does not mean coefficientwise nonnegativity.

The path interpretation also gives the formal Stieltjes continued fraction

$$\sum_{n \ge 0} \mu_n(\alpha) z^n = \cfrac{1}{1 - \cfrac{\alpha_1 z}{1 - \cfrac{\alpha_2 z}{1 - \cfrac{\alpha_3 z}{\ddots}}}}. \tag{15}$$

Indeed, if F_h enumerates excursions above height h, the first-return decomposition gives $F_h = 1 + \alpha_{h+1} z F_{h+1} F_h$. The denominator in $F_h = (1 - \alpha_{h+1} z F_{h+1})^{-1}$ has constant term 1. Moreover, $[z^n] F_0$ only uses heights up to n, so the infinite fraction is well-defined coefficient by coefficient. There is no analytic convergence or sign condition.

5 Exact finite checks and scope

The accompanying `verify.py` provides three exact checks. It generates the path polynomials by stepwise dynamic programming and substitutes them into the original negative Toeplitz minors (1), checking all symbolic identities for $1 \le i \le 10$. It also checks the defining equations through $i = 40$ for each of the three integer assignments

$$y_j = 1, \qquad y_j = j, \qquad y_j = (-1)^j ((j \bmod 5) + 1).$$

Finally, it compares the path polynomials with the composition formula through μ_{12}, equivalently ℓ_{13}. The recorded output is `verification.json`; the calculation uses exact integer and polynomial arithmetic. The last comparison does not claim a symbolic Toeplitz determinant check through index 13.

These finite checks aid reproducibility and detect indexing or sign errors. The all-index statement follows from the preceding proof, not from extrapolation of the checks. The argument establishes polynomiality and the displayed coefficient formula; it does not establish the SAGBI or initial-algebra conjectures of [1].

AI assistance. This work was developed with AI assistance. The mathematical argument is given in full, and exact finite verification code is supplied for reproducibility.

References

[1] Lukas Katthän, *The Lecture Hall Cone as a toric deformation*, arXiv:1809.01377v1 (2018). `https://arxiv.org/abs/1809.01377v1`.

[2] Alan D. Sokal, *A simple algorithm for expanding a power series as a continued fraction*, Expositiones Mathematicae **41** (2023), 245–287. doi:10.1016/j.exmath.2022.12.001. Also arXiv:2206.15434.

[3] Philippe Flajolet, *Combinatorial aspects of continued fractions*, Discrete Mathematics **32** (1980), 125–161. `https://algo.inria.fr/flajolet/Publications/Flajolet80b.pdf`.